Experimental probability

1 A retail store served 773 customers in October, and there were 44 complaints during that month.

Calculate the experimental probability that a customer complains. Give your answer as a percentage, rounded to the nearest whole percent.

2 Find the experimental probability of the following:

a Flipping heads from a coin that was flipped 184 times with 93 heads recorded.

b A computer being faulty at a factory where 1000 computers were tested and 15 were found to be faulty.

c A car driving by is white, given that 1919 cars had gone by and 77 of them were white.

3 At a set of traffic lights, the green light is on for 24 seconds, then the yellow light lasts for 3 seconds, and then the red light is on for 13 seconds. This cycle then repeats continuously.

If a car approaches the traffic light, calculate the probability that the light will be:

a Green b Yellow c Red

4 An insurance company found that in the past year, of the 2558 claims made, and 1493 of them were from drivers under the age of 25. Calculate the experimental probability, as a whole percentage, that a claim is filed by someone:

a Under the age of 25. b 25 years or older.

5 The experimental probability that a person uses public transport is 50%. Out of 500 people, how many would you expect to use public transport?

6 Hermione rolled a standard six-sided die 60 times.

a State the theoretical probability of rolling a six.

b How many times would she expect a six to appear?

c After she finished rolling the die, she noticed that she had rolled a six 24 times. Find the experimental probability of getting a six.

d Was the theoretical and experimental probability of getting a six the same? Explain your answer.

7 A factory produces tablet computers. In March it produced 8000 tablets, and 240 were found to be faulty.

a Calculate the experimental probability that a tablet produced by the factory is faulty.

b The factory plans to produce 9000 tabl[illegible] April. How many should they expect to be faulty?

8 Uther flipped a coin 14 times.

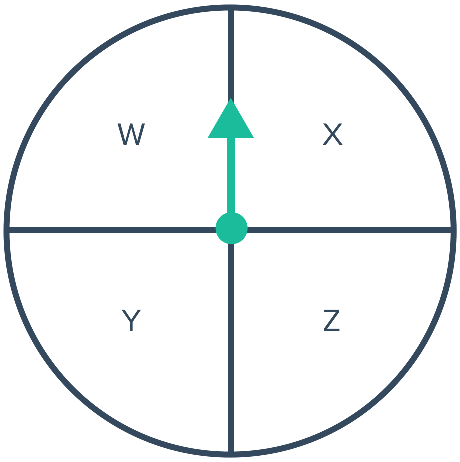
a How many times would he expect a tails to appear?

b After flipping the coins, he noticed that tails had appeared 4 times. Find the experimental probability of getting a tails.

9 Derek spun the following spinner 20 times:

a How many times would he expect the arrow to land on X?

b After he finished spinning, he noticed that the arrow fell on X 8 times. Find the experimental probability of getting an X.



10 Oprah has a bag with 2 red balls, 2 blue balls, and 2 green balls in it. She took a ball out of the bag and returned it 24 times.

a How many times would she expect to get a green ball?

b After she finished, she noticed that she had drew a green ball 18 times. Find the experimental probability of getting a green ball.

11 Georgia is drawing a card out of a deck of 10 cards, labeled from 1 to 10. She drew a card and returned it to the bag 40 times.

a How many times would she expect to get the card with 6 on it?

b After she finished, she noticed that the card with 6 on it was drawn out of the bag 21 times. Find the experimental probability of getting a card with 6 on it.

Tables

12 To prepare for the week ahead, a restaurant keeps a record of the number of each main meal ordered throughout the previous week.

a How many meals were ordered altogether?

b Calculate the experimental probability that a customer will order a beef meal.

Give your answer as a percentage, rounded to the nearest whole percent.

Meal	Frequency
Chicken	54
Beef	32
Lamb	26
Vegetarian	45

13 Five schools compete in a basketball competition. The results from the last season are given in the table below:

Schools competing each game	Winner
St Trinian's vs Ackley Bridge	St Trinian's
Ackley Bridge vs Summer Heights	Summer Heights
Lakehurst vs Marquess	Marquess
Marquess vs St Trinian's	St Trinian's
St Trinian's vs Summer Heights	Summer Heights
St Trinian's vs Lakehurst	Lakehurst
Ackley Bridge vs Marquess	Ackley Bridge
Marquess vs Summer Heights	Summer Heights
Lakehurst vs Summer Heights	Lakehurst
Ackley Bridge vs Lakehurst	Lakehurst

Calculate the experimental probability that Ackley Bridge wins one of their matches.

14 The table shows the number of trains that arrived on time at the local station during the week:

a Calculate the experimental probability that a train will be on time on Monday, as a whole percentage.

b Which day had the highest experimental probability of a train being on time?

c Calculate the experimental probability of a train arriving on time across the entire week, as a whole percentage.

Day	Number of trains	On time
Monday	29	22
Tuesday	24	23
Wednesday	21	19
Thursday	27	22
Friday	23	20

15 The following table shows the outcomes of tossing three coins 102 times:

Find the experimental probability of:

a Tossing 3 tails.

b Tossing at least 2 heads.

c Tossing at least 1 tail.

d Tossing only 1 head.

e Tossing exactly 2 tails.

Outcome	Frequency
HHH	11
HHT	12
HTH	11
HTT	16
THH	12
THT	15
TTH	10
TTT	15

16 Boxes of toothpicks are examined and the number of toothpicks in each box is recorded in the table shown:

If the number of toothpicks of another box were counted, find the experimental probability it will:

a Contain 89 toothpicks.

b Contain more than 90 toothpicks.

c Contain less than 90 toothpicks.

Number of toothpicks	Number of Boxes
87	0
88	6
89	4
90	1
91	1
92	2
93	1

17 High school students attending an international conference were asked to register what language they speak other than English. The results are shown in the table below:

a How many students attended the conference?

b Find the probability that a student chosen at random speaks:

i French

ii Mandarin

iii Arabic or Spanish.

iv Spanish or Other.

Language	Frequency
French	20
Arabic	13
Spanish	21
Mandarin	19
Other	37

18 The table shows the results of rolling a die: 

Outcome	Frequency
1	13
2	12
3	20
4	14
5	15
6	16

a How many times was the die rolled?

b What is the experimental probability of rolling a 2?

c What is the experimental probability of rolling an even number?

19 The table shows the fighting style of each competitor in a mixed martial arts tournament fought last year:

Event	Frequency
Karate	40
Wrestling	54
Judo	47
Taekwondo	59

a Find the total number of competitors.

b Find the experimental probability of a wrestler winning.

c Find the experimental probability of a Taekwondo fighter not winning.

d If 100 more competitors joined the competition, how many of them would you expect to use Karate as their fighting style?